

WELLSENSE · REQUIREMENTS

WellSense requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/wellsense		
Source	docs/03-requirements.md @ 708a938		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

WellSense requirements

These are first-pass requirements for the concept. Targets are proposals for review, to be revised after co-design and checked by calculation at TRL 3. The status column states whether the concept in WLS-PRC-001 meets each target on first-order estimates.

Table 1. Requirements and status at TRL 2.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Measurement range (water above the probe)	0 to 10 m, with a 0 to 20 m option	Datasheet	Met by the proposed transducer
R2	Installation depth below the wellhead	Probe to 60 m or more	Cable and tube length; hanger load	Met by design (cable sized per well)
R3	Resolution	1 mm or better	ADC and shunt calculation	Met: about 0.5 mm (estimate)
R4	Accuracy after field calibration against a manual tape	±20 mm over the range	Calculation, then side-by-side with a manual tape	Not met on datasheet: about ±50 mm at 0.5 % of full scale; may be met with a 0.25 % sensor or two-point field calibration, unverified
R5	Long-term drift	20 mm a year or less, checked by a manual reading each quarter	Field record	Unverified, at risk: drift of low-cost transducers is not published
R6	Reading interval	15 min default; 1 min during a pumping test	Firmware configuration	Met by design
R7	Local storage	1 year of 15 min readings without a link	Storage calculation	Met: about 0.3 MB a year against FieldNode flash (estimate)
R8	Energy	Probe and interface use 5 % or less of the FieldNode sensor energy budget at 15 min	Power budget	Met: about 18 mWh a day, about 0.6 % (estimate)
R9	Water safety	All wetted parts 316 stainless steel or materials certified for drinking water contact; well sealed against surface water	Material certificates	Not met on evidence: cable jacket and sensor seal certification for low-cost transducers unknown
R10	Compatibility with the existing pump	Probe in its own access tube; no contact with pump, riser or pump cable	Design review, installation trial	At risk: fitting a tube may need the pump pulled; no room in narrow casings
R11	Installation	Two trained people, hand tools, 2 h or less when an access tube already exists	Installation trial	Unverified
R12	Cost	WellSense parts \$180 or less per well (FieldNode core counted separately)	Priced BOM	Met: about \$170; not met if FieldNode is included (about \$296)
R13	Data use	Dashboard shows level below ground, daily drawdown and change against the same month last year on a basic phone; CSV export	Demonstration with sample data	Unverified (software not started)
R14	Data ownership and privacy		Design review	Met by design

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
		Water level, time and well ID only; the hosting community controls sharing		
R15	Environment	Probe IP68 at 1.5 times range; above-ground parts IP65, -10 to 55 °C	Datasheets, later field test	Met on datasheets; unverified in the field
R16	Tamper resistance	Wellhead parts lockable; no exposed cable at reachable height	Design review	At risk: cable run from the wellhead to the post is exposed in the concept

■ Assumptions

- Water density 1,000 kg/m³; 1 m of water is 9.81 kPa. Temperature changes density by about 0.3 % between 4 and 25 °C, which firmware corrects using the probe temperature where available.
- A vented (gauge) transducer is assumed, so the reading is already referenced to air pressure. The barometric sensor is kept to record the aquifer's response to air pressure and to detect a blocked vent.
- Accuracy stated as a percentage of full scale applies over the whole 10 m range; ±0.5 % of 10 m is ±50 mm.
- Sensor supply 12 V, current up to about 22 mA, powered for about 2 s per reading, from FieldNode's switched 12 V rail at about 80 % boost efficiency.